

OSSS-2

Question:

1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

Solution:

```
GNU nano 8.7.1 myshell.c
#include <stdio.h>
#include <string.h>

int main()
{
    char command[100];

    while(1)
    {
        printf("myshell> ");

        fgets(command,sizeof(command),stdin);

        command[strcspn(command,"\n")] = 0;

        if(strcmp(command,"exit")==0)
        {
            printf("Exiting Shell\n");
            break;
        }

        printf("You entered: %s\n",command);
    }

    return 0;
}

ubuntu@ubuntu:~/skill2$ nano myshell.c
ubuntu@ubuntu:~/skill2$ gcc myshell.c -o myshell
ubuntu@ubuntu:~/skill2$ ./myshell
myshell> Hello World...!
You entered: Hello World...!
myshell> This is Operating System
You entered: This is Operating System
myshell> This is skill2
You entered: This is skill2
myshell> exit
Exiting Shell

ubuntu@ubuntu:~/skill2$ nano flow.txt
GNU nano 8.7.1 flow.txt
Start
|
Display Prompt
|
Read User Input
|
Is Command = exit ?
|
Yes -----> End
|
No
|
Display Command
|
Return to Prompt
```